Live-well^{TM} OsteSAMIN^{TM} *Plus* CHONDROITIN & MSM Powder

Each 5.8g sachet contains:	
Glucosamine Sulphate Potassium Chloride	
equiv. to Glucosamine Sulphate	1500mg
Chondroitin Sulphate Sodium	1200mg
equiv. to Chondroitin Sulphate 1078mg	
MSM (Methylsulfonylmethane)	1000mg

DESCRIPTION:
Powder: A yellow powder with odour of orange flavour
Solution: A yellow solution with odour of orange flavour.

PHARMACODYNAMICS:
Glucosamine plays a major role in the promotion and maintenance of the structure and function of cartilage in the bone joint. Glucosamine stimulates the biosynthesis of mucopolysaccharides which is the key structural components of joint cartilage which are essential for proper joint function. Glucosamine also increases synovial fluid production and viscosity, giving better lubricant activity for the joints. Glucosamine sulphate is well-absorbed after oral intake, and is distributed mainly to bones and joint tissues.

Chondroitin sulfate is vital for the structure and function of articular cartilage. Chondroitin sulfate is an important component of aggrecan which is found in articular cartilage. Aggrecan confers upon articular cartilage shock-absorbing properties. It does this by providing cartilage with a swelling pressure that is restrained by the tensile force of collagen fibers. This balance confers upon articular cartilage the deformable resilience vital to its function.

Methylsulfonylmethane (MSM) is an organic sulphur-containing compound. It is an oxidation product of the organic solvent, dimethyl sulfoxide (DMSO). DMSO has been used in the treatment of arthritis and connective tissue injuries because cartilage has a high content of sulphur, and sulphur is needed for the formation of connective tissue, MSM as a source of sulphur could be useful in the management of conditions such as osteoarthritis and joint injuries where there is degeneration or destruction of cartilage.

PHARMACOKINETICS:
Glucosamine:
Absorption
After oral administration, bioavailability is low due to first-pass hepatic metabolism 26% The gastrointestinal absorption is close to 90%.
Distribution
Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins).
Volume of Distribution: 2.5 Liters
Metabolism
Liver, extensive

The first-pass effect in the liver in which more than 70% of glucosamine is metabolized.

Excretion

Renal Excretion, 10%

Feces, 11%

Part of a dose of glucosamine sulfate is eliminated as carbon dioxide via expired air.

Chondroitin sulphate:

Absorption

After oral administration, the bioavailability is 15-24%. A 10% of absorbed chondroitin sulphate appears in unmetabolized form and 90% as depolymerized derivatives with lower molecular weight, as a result of first-pass effect metabolism.

Distribution

The distribution volume of chondroitin sulphate is relatively low. In humans, it has shown to have affinity for joint tissues. At least 90% of chondroitin sulphate dose is first metabolized by lysosomal sulfateses, and after that it is depolymerized by hyaluronidases ie, 13-glucuronidases and 13-N-acetyl hexose amidases. Liver, kidneys and other organs are involved in the depolymerization of chondroitin sulphate. Chondroitin sulphate remains in the body for 5-15 hours.

Excretion

Main part of chondroitin sulphate and its depolymerized derivatives are eliminated via the kidneys.

Methylsulfonylmethane:

Soft tissue distribution of radioactivity indicated a fairly homogeneous distribution throughout the body with relatively lower concentrations in skin and bone. Approximately 85.8% of the dose was recovered in the urine after 120 hours, whereas only 3% was found in the feces. No quantifiable levels of radioactivity were found in any tissues after 120 hours, indicating complete elimination of MSM. This indicates that MSM is rapidly absorbed, well distributed, and completely excreted from the body.

INDICATION:

Adjunctive therapy for osteoarthritis

ROUTE OF ADMINISTARTION:

Oral administration.

RECOMMENDED DOSE:

Dosage: One sachet a day.

Direction for use:

Mix 1 sachet powder into a glass of water (approximately 200ml). Stir until dissolved.

Sachets should be taken 15 minutes before meals.

CONTRAINDICATIONS:

Contraindicated in individuals who have hypersensitivity to shellfish, Glucosamine, Chondroitin sulfate and Methylsulfonylmethane (MSM).

WARNING AND PRECAUTION:

Those with type 2 diabetes and those who are overweight and have problems with glucose tolerance should have their blood sugars carefully monitored if they consume glucosamine.

Glucosamine treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2-3 weeks. Therefore, it is advisable to take an analgesic/anti-inflammatory drug if required during the first 2-3 weeks of therapy with glucosamine.

Administration during the first three months of pregnancy must be avoided.

Safety and effectiveness have not been established in children therefore children, should avoid using glucosamine

The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision.

Derived from Seafood, therefore should not be given to patients who are allergic to shellfish.

A doctor should be consulted in order to exclude the presence of other joint conditions / disesases for which an alternative treatment should be considered

Effects on Ability to Drive and Use Machines
No effects on the ability to drive or to operate machines are expected.

INTERACTIONS WITH OTHER MEDICAMENTS:
Effects on glucose metabolism & antidiabetic agents:
It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake.

Glucosamine may increase insulin resistance and consequently affect glucose tolerance.

It may reduce antidiabetic agent effectiveness eg when used with these antidiabetic agent :

Acarbose, Acetohexamide, Chlorpropamide, Glipizidede, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride , Tolbutamide, Troglitazone,

Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely/more frequently

Reduced effectiveness when used with glucosamine: Doxorubicin , Etoposide , Teniposide
Warfarin

- elevations of International Normalized Ratio serum values and potentiation of anticoagulant effects

- If concomitant therapy is necessary, the patient's INR should be more closely monitored

PREGNANCY AND LACTATION:
If pregnant or breastfeeding, please consult doctor or health care professional before taking this product.

SIDE EFFECTS:
Cardiovascular:
Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.
Central nervous system:
Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).
Gastrointestinal:
Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.
Skin:
Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

OVERDOSAGE:
In the event of an acute overdose, the patient should be observed closely and symptomatic treatment should be instituted.

STORAGE:
Store in a dry place below 30°C.

SHELF-LIFE:
3 years

PRESENTATION:
Box of 10, 15, 20, or 30 sachets.

MANUFACTURED BY:
NORIPHARMA SDN. BHD. (792633-A)
Lot 5030, Jalan Teratai, 5
1/2 Miles off Jalan Meru,
41050 Klang, Selangor,
Malaysia.

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